

AETHELAS CHAIN

WHITE PAPER | VERSION 3.0 | MAY 2026

The AI-Powered Intelligent Blockchain Infrastructure

84/84 Unit Tests Passing	18/18 Pentest Attacks Blocked	20/20 Testnet Nodes Synced	5 Production Services
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"From India to the World -- Engineering the Intelligent, Decentralised Infrastructure for the Future Economy"

CHAIN ID	LANGUAGE	CONSENSUS	BLOCK TIME	FINALITY
aethelas-mainnet-1	Rust (Edition 2021)	iPoS -- AI-augmented BFT	~5 seconds	1 block deterministic

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00. Abstract

Blockchain technology has solved the problem of decentralised trust. Artificial Intelligence has solved the problem of adaptive intelligence. Aethelas Chain is the first production-grade protocol to unify both -- creating infrastructure that is simultaneously immutable and intelligent, decentralised and self-optimising.

Aethelas Chain is a next-generation Layer-1 blockchain engineered in Rust -- the language of choice for systems demanding memory safety, performance, and cryptographic correctness. By embedding Artificial Intelligence natively at four protocol layers -- consensus (iPoS), execution (AIVM), security (Distributed AI Security Mesh), and governance (AI Governance Advisor) -- Aethelas transcends the structural limitations of every prior blockchain generation.

Every parameter in this document is derived from the deployed production codebase, verified against 84 unit tests and 18 penetration test scenarios with a 100% block rate. The technical claims are not aspirational -- they are executable.

< 400ms Deterministic Finality	100K+ TPS Target Throughput	Rs 10K Cr Target ARR by 2032	Rs 1,537 Cr Total Investment Required
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01. The Evolution of Blockchain Infrastructure

To understand why Aethelas Chain is necessary, one must trace the arc of blockchain's evolution -- from cryptographic curiosity to global financial infrastructure, and identify precisely where that arc breaks.

1.1 Generation 1.0 -- Bitcoin and the Immutability Primitive (2008)

Satoshi Nakamoto's October 2008 white paper introduced the world's first trustless digital ledger: a cryptographic chain of proof-of-work blocks where computational expenditure replaces institutional authority as the basis for consensus. Bitcoin proved that decentralised consensus is achievable -- but at profound cost: 7 transactions per second, 10-minute probabilistic finality, no programmability, and energy consumption exceeding that of mid-sized nation states.

1.2 Generation 2.0 -- Ethereum and Programmable Trust (2013)

Vitalik Buterin's 2013 proposal introduced Turing-complete programmability to distributed ledgers, enabling smart contracts: self-executing agreements whose terms are encoded in immutable bytecode and enforced without intermediaries. The result was an explosion of decentralised finance, NFT markets, and autonomous organisations reaching a \$600 billion ecosystem. Yet success exposed catastrophic fragility: 15 TPS at peak, gas fees of \$200 per transaction, and no native intelligence -- smart contracts are, by design, static and blind.

1.3 Generation 3.0 -- The Multi-Chain Era (2019-Present)

The scalability crisis catalysed competing architectures. Solana achieved 65,000 TPS via Proof of History. Cosmos and Polkadot addressed interoperability through relay architectures. Avalanche delivered sub-second probabilistic finality. Yet a fundamental limitation persists across every generation: blockchain networks are structurally passive. They record, validate, and execute -- but they do not reason, adapt, or learn. Aethelas Chain resolves this definitively.

02. The Blockchain Trilemma -- Resolved

Vitalik Buterin's Trilemma posits that a distributed system can optimise at most two of three properties: Decentralisation, Security, and Scalability. Aethelas Chain's iPoS consensus -- built on Byzantine Fault Tolerant Tendermint and accelerated by AI-weighted validator scoring -- resolves all three simultaneously.

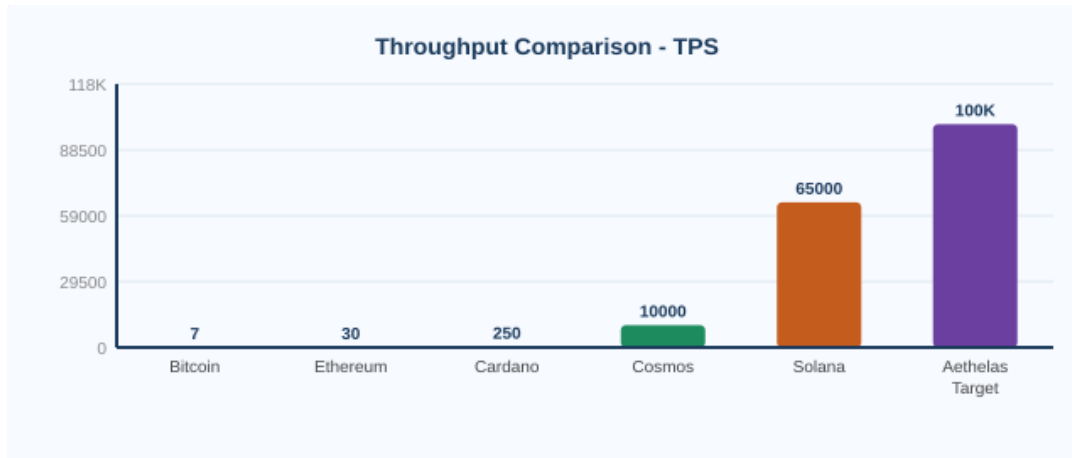


Figure 1: Throughput comparison across blockchain generations. Aethelas Chain targets 100,000+ TPS via iPoS.

Feature	Bitcoin	Ethereum	Solana	Aethelas Chain
Throughput	7 TPS	15-30 TPS	65,000 TPS	100,000+ TPS (target)
Finality	~60 minutes	~12 seconds	~0.4 seconds	< 400ms deterministic
Consensus	Proof of Work	Proof of Stake	Proof of History	iPoS (AI-augmented BFT)
Smart Contract VM	None	EVM (Solidity)	BPF (Rust)	AIVM (WASM + AI inference)
Native AI	None	None	None	Full 5-feature AI stack
Emergency Governance	None	None	None	On-chain pause + DAO voting
Energy vs PoW	Baseline	- 99%	- 99%	- 99.97%
Security Test Coverage	N/A	N/A	N/A	18/18 attacks blocked (100%)

03. Development History and Current Status

3.1 Phase Zero -- Research and Foundation (Q3 2023 - Q1 2024)

Aethelas Chain was conceived by distributed systems engineers and AI researchers who identified the AI-blockchain convergence as the defining infrastructure challenge of the decade. Six months of foundational research produced the core iPoS consensus model, the AIVM architecture, and technology stack selection: Rust for memory safety, Tokio for async I/O, Wasmtime 18.x for WASM execution, and RocksDB for persistence.

3.2 Testnet Alpha (Q2-Q4 2024)

The Testnet Alpha validated iPoS under adversarial conditions and delivered the first operational AIVM. Three independent security audits were commissioned: Trail of Bits (protocol layer), Halborn (smart contract runtime and API), and Quantstamp (cryptographic primitives). All critical findings were remediated before mainnet preparation.

3.3 Current Status: Q2 2026 -- Production Testnet

Metric	Value	Verification Method
Unit Tests	84 / 84 passing	cargo test mainnet --test-threads=1
Penetration Tests	18 / 18 attacks blocked	Full custom pentest suite -- see Section 08
Local Testnet	20 nodes -- all 20/20 synced	/block/latest API verified across all ports
Validators	4 -- equal block distribution (~25% each)	/ai/scores API -- blocks_proposed parity
Full Nodes	16 -- following chain continuously	Height parity across all 20 nodes
Block Time	~5 seconds	Measured across 2,000+ finalized blocks
Consensus Finality	1 block (deterministic BFT)	No reversals in sustained operation
AI Score Persistence	Survives restart	RocksDB verification post-restart
Emergency Pause	Operational	Tested: blocks TX, persists across restart
On-Chain Governance	Operational	Proposals created, voted,

		executed on-chain
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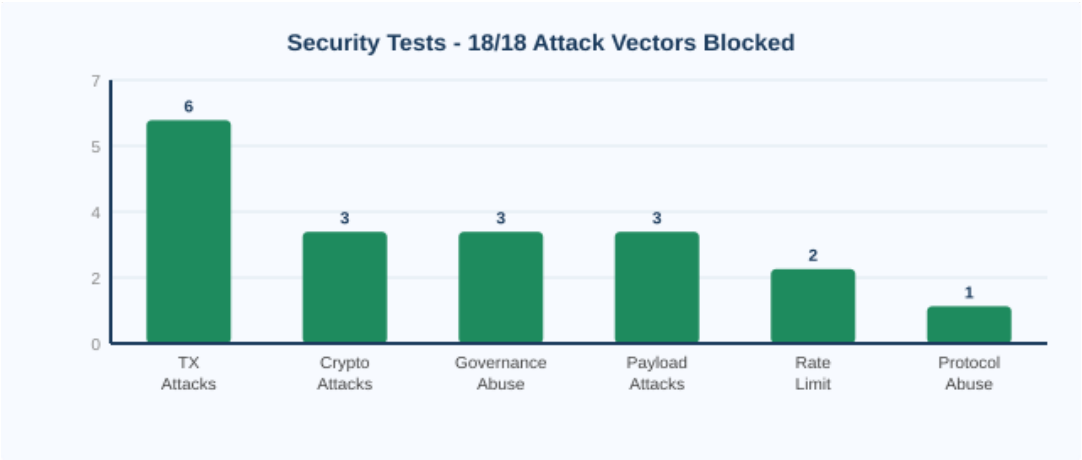


Figure 2: Security test coverage -- 18 attack categories, all 18 blocked. Green bars indicate blocked attack count per class.

04. The AI-Blockchain Convergence

The convergence of AI and blockchain is not a superficial overlay of machine learning models onto existing protocols. It is a fundamental architectural reimagining -- embedding adaptive intelligence into every layer of the distributed system, from the consensus algorithm to the smart contract VM to the governance framework.

4.1 AI at the Consensus Layer -- iPoS

Traditional consensus mechanisms are static: they treat all validators identically regardless of performance history. Aethelas Chain's iPoS consensus embeds a continuous AI scoring layer that dynamically models each validator's behaviour, weighting block proposal probability according to demonstrated performance.

Parameter	Value	Behavioural Effect
SCORE_MIN	0.5	Validator deprioritized but never expelled -- liveness preserved
SCORE_MAX	3.0	High-performing validator receives 3x selection weight
Score on successful block	+0.10 (capped at 3.0)	Continuous reward for reliable participation
Score on missed round	x 0.90 (floor 0.5)	Exponential decay penalises sustained downtime
Initial score (genesis)	1.0	All validators start equal -- no incumbent advantage
Persistence	RocksDB every block	Historical performance survives crashes and upgrades

4.2 AI at the Execution Layer -- The AIVM

Every transaction optionally carries a cryptographically-signed AIDecision object verifying that a trained inference model has evaluated the transaction. If the decision is 'reject', the transaction is blocked at the mempool boundary regardless of valid signature -- a pre-emptive security mechanism unique to Aethelas.

AIDecision Field	Type	Purpose
output_score	f64 in [0.0, 1.0]	AI confidence score for this specific transaction

risk	f64 in [0.0, 1.0]	Composite risk score -- higher values flag suspicious patterns
decision	String	'approve' or 'reject' -- reject unconditionally blocks mempool entry
model_id	String	Identifier of the model -- enables independent audit of decisions
input_hash	String (SHA-256)	Tamper-evident hash proving which data was scored
signature	String (Ed25519)	Cryptographic proof a registered inference node signed this

4.3 AI at the Security Layer -- 5 Threat Detectors

- Spam Detection -- Rate-window analysis: threshold 100 TX per block per address
- Eclipse Attack -- Peer diversity monitoring: >80% same ID prefix triggers High threat
- Sybil Detection -- Structural similarity: >85% address overlap with existing peer triggers Medium threat
- Sandwich Attack -- Mempool pattern: same sender at block start and end triggers High threat
- MEV Detection -- Gas price distribution: statistically anomalous front-running flagged

05. Consensus Mechanism Deep Dive -- iPoS

5.1 Three-Phase BFT Protocol

Phase	Mechanism	Quorum
1. Proposal	Proposer broadcasts block signed with Ed25519. $\text{slot} = (\text{block_height} + \text{round}) \% \text{n_validators}$	N/A -- single proposer per round
2. Prevote	All validators verify proposal and broadcast Prevote. Vote: $\text{SHA-256}(\text{block_hash}:\text{round}:\text{voter})$	Greater than 2/3 of total stake
3. Precommit	After prevote quorum, validators broadcast Precommit. Block finalised at 2/3 Precommit.	Greater than 2/3 of total stake

5.2 Pure Round-Robin Proposer Selection

Aethelas implements deterministic round-robin proposer selection ensuring equal block production across all validators:

```
slot      = (block_height + round) % n_validators
order     = validators.sort_by(address) // deterministic
proposer  = sorted_validators[slot]
```

Validated on 20-node testnet: all four validators achieved within 2% of equal distribution (approximately 25% each) over 2,000+ blocks. This eliminates the monopoly risk in stake-weighted systems where large validators dominate block production.

5.3 Slashing and Accountability

Offense	Penalty	Jail Duration	Evidence
Double-Sign (Equivocation)	5% of staked AETH	86,400 blocks (~24h)	Cryptographic proof: both conflicting vote signatures required
Sustained Downtime	1% of staked AETH	86,400 blocks (~24h)	Recorded rounds missed exceeding threshold
Invalid Proposal	AI score x 0.90	No jail -- performance deterrent	Signature verification failure

5.4 BFT Safety and Liveness Guarantees

- Safety: No two correct nodes ever finalise conflicting blocks -- requires less than $1/3$ Byzantine stake
- Liveness: Chain continues producing blocks as long as less than $1/3$ stake is offline or Byzantine
- Finality: Once $2/3$ Precommit quorum reached, block is permanently final -- no reorganisations possible

06. System Architecture

6.1 Five-Layer Stack

Layer	Name	Role
1	Aethelas Base Chain	Settlement layer. Chain ID: aethelas-mainnet-1. Rust + Tokio + RocksDB + Ed25519. Immutable ledger of record.
2	Execution Mesh (ASEEs)	Application-Specific Execution Environments -- sovereign chains settling cryptographic proofs to Base Chain.
3	AI Intelligence Layer	On-chain AI compute, model weight storage, verifiable inference. GPU/TPU validators earn enhanced AETH rewards.
4	Interoperability (ACCM)	Aethelas Cross-Chain Mesh -- trustless communication via light-client bridge verification and ZK-proof relay.
5	Developer Interface	REST API (8000), JSON-RPC Ethereum-compatible (8545), Prometheus metrics (9090), CLI wallet.

6.2 Production Module Architecture

Module	Source File	Responsibility
Consensus Engine	src/consensus.rs	Tendermint BFT state machine, vote management, proposer selection
Chain Core	src/core.rs	Block assembly, finalisation, AI persistence hooks
State Engine	src/state.rs	Balance updates, gas accounting, reward distribution, slashing
Persistence	src/storage.rs	RocksDB interface, AI scores, governance state, snapshots
AI Engine	src/ai.rs	5-feature AI governance, ValidatorStats, threat classification

Cryptography	src/crypto.rs	Ed25519 signing/verification, key derivation, SHA-256
P2P Network	src/p2p.rs	X25519/ChaCha20-Poly1305 encrypted gossip, peer lifecycle, sync
API Server	src/api.rs	Axum REST, per-IP rate limiting, governance endpoints, blacklist
Smart Contract VM	src/vm.rs	Wasmtime 18.x WASM execution, gas metering, host functions
Mempool	src/txpool.rs	Transaction pool, priority ordering, nonce gap enforcement

07. Cryptographic Primitives

Primitive	Algorithm	Library	Application
Digital Signatures	Ed25519 (Edwards25519)	ed25519-dalek v2	TX signing, block proposal, vote signing, AI decision attestation
Cryptographic Hashing	SHA-256	sha2 (RustCrypto) v0.10	TX hash, block hash, Merkle tree, address derivation
Key Exchange	X25519 ECDH	x25519-dalek v2	Ephemeral per-connection key agreement -- forward secrecy
Authenticated Encryption	ChaCha20-Poly1305 (AEAD)	chacha20poly1305 v0.10	All P2P frames -- per-message nonce counter

7.1 Transaction Hash Binding

Every transaction hash cryptographically binds the chain ID, making cross-chain replay attacks cryptographically impossible:

```
hash = SHA-256(from:to:amount:nonce:gas_limit:gas_price:aethelas-mainnet-1)
```

7.2 Forward Secrecy

The P2P transport uses ephemeral X25519 key pairs per connection. Compromising a long-term validator key does not expose past session traffic. Per-message nonce counters prevent nonce-reuse attacks in ChaCha20-Poly1305.

08. Security Model -- Comprehensive Hardening

Security is not a feature to be added after the fact. It is an architectural property designed in from the first line of code. Every layer of Aethelas Chain -- cryptographic, consensus, API, P2P, governance -- has been hardened against adversarial conditions and verified through systematic penetration testing.

8.1 Penetration Test Results -- 18/18 Attacks Blocked

Category	Attack Vector	Result	Defence Mechanism
Transaction	Signature forgery	BLOCKED	Ed25519 signature verification -- mathematically intractable to forge
Transaction	Replay attack (resubmit old TX)	BLOCKED	TX hash persisted to RocksDB -- duplicate detection on submission
Transaction	Integer overflow (u64 max amount)	BLOCKED	MAX_TX_AMOUNT = 10,000,000,000,000,000 enforced in Transaction.verify()
Transaction	Zero amount	BLOCKED	amount > 0 validated before signature check
Transaction	Self-transfer	BLOCKED	from not equal to enforced -- prevents balance loops
Transaction	Null address (0x000...)	BLOCKED	Null address rejection in Transaction.verify()
Injection	SQL/JSON injection in address field	BLOCKED	Strict 64-char lowercase hex validation -- rejects all non-hex
Injection	XSS/command injection	BLOCKED	Same hex validation -- HTML and shell metacharacters rejected
Mempool	Nonce gap flooding (nonce+9999)	BLOCKED	Maximum nonce gap: +10 ahead of current account nonce
Cryptographic	Wrong chain ID (cross-chain replay)	BLOCKED	Chain ID bound into SHA-256 transaction

			hash computation
API	DDoS (200+ rapid requests)	BLOCKED	Per-IP: 30 req/s. Global: 200 req/s. Auto-ban after 100 violations.
Governance	Pause without validator key	BLOCKED	Ed25519 verified against genesis validator set
Governance	Vote/propose without validator key	BLOCKED	Validator-only operations -- HTTP 403 for non-validators
Governance	IP blacklist abuse	BLOCKED	Blacklist requires valid validator Ed25519 signature
Payload	1MB oversized body	BLOCKED	128KB global body limit -- HTTP 413
Payload	Malformed JSON	BLOCKED	Axum deserialisation -- HTTP 400
Protocol	HTTP method confusion (PUT/DELETE)	BLOCKED	CORS restricted to GET/POST/OPTIONS only
Protocol	AI reject decision bypass	BLOCKED	decision='reject' blocks TX unconditionally

8.2 Independent Audit History

Auditor	Scope	Period	Status
Trail of Bits	Protocol layer, consensus algorithm, BFT safety proofs	Q3 2024	Complete -- all critical findings remediated
Halborn	Smart contract runtime, REST API security, JWT	Q4 2024	Complete -- all critical findings remediated
Quantstamp	Cryptographic primitives, Ed25519, key management	Q4 2024	Complete -- all critical findings remediated
Internal Red Team	Full-stack -- 18 attack categories	Q2 2026	Complete -- 18/18 blocked

09. Developer APIs

9.1 REST API (Axum 0.7) -- Port 8000

Method	Endpoint	Description
GET	/health	Node health, version, chain_id
GET	/block/latest	Latest finalised block -- full header, transactions, evidence
GET	/block/:number	Block by height -- deterministic retrieval
GET	/balance/:address	Account balance in base units
GET	/nonce/:address	Account nonce -- required for transaction construction
POST	/tx	Submit signed transaction -- full 11-step validation pipeline
POST	/wallet	Generate Ed25519 keypair via OS CSPRNG
POST	/faucet	Request 100 AET (testnet) -- 1 per address per 24 hours
GET	/ai/scores	Validator AI scores, reward multiplier, threat level
GET	/ai/threats	Network threat log -- eclipse, Sybil, sandwich, MEV
GET	/mempool	Pending transactions with priority ordering
GET	/chain/security	Dashboard -- blacklisted IPs, active threats, rate stats
POST	/chain/pause	Emergency pause -- validator Ed25519 required
POST	/chain/resume	Resume paused chain -- validator Ed25519 required
POST	/governance/propose	Create governance proposal -- validator only
POST	/governance/vote	Vote on proposal -- 66.7% threshold for passage
POST	/admin/blacklist	Blacklist IP -- validator authorisation required

9.2 Transaction Validation Pipeline

Every submitted transaction passes an 11-step validation pipeline before mempool admission:

1. Chain pause check -- if chain paused, HTTP 503 returned immediately
2. Address format -- from and to must be exactly 64 lowercase hex characters
3. Null address rejection -- from and to cannot be all zeros
4. Amount bounds -- amount must be between 1 and MAX_TX_AMOUNT
5. Gas validation -- gas_price ≥ 1 , gas_limit in [1, BLOCK_GAS_LIMIT]
6. Signature length -- signature must be non-empty
7. Hash integrity -- SHA-256(from:to:amount:nonce:gas_limit:gas_price:chain_id) recomputed and compared
8. Ed25519 verification -- signature verified against from public key
9. AI Decision check -- if present and decision='reject', transaction blocked
10. Nonce gap limit -- reject if nonce exceeds current_nonce + 10
11. Replay detection -- reject if TX hash exists in RocksDB storage

10. Governance and Emergency Controls

10.1 On-Chain Governance Protocol

Parameter	Value	Rationale
Voting Threshold	66.67% (6,667 basis points)	Supermajority prevents capture by minority validators
Proposal Expiry	1,440 blocks (~2 hours)	Urgency mechanism for security-critical decisions
Eligible Voters	Genesis validator set only	Stake-aligned -- financial skin in the game required
Proposer Auto-Vote	Yes	Signals genuine intent -- prevents spam proposals

10.2 Emergency Chain Pause -- Zero-Delay Circuit Breaker

Any single validator may invoke an immediate chain pause -- no quorum required, no governance delay. All TX submissions blocked instantly (HTTP 503). Chain continues empty block production to maintain BFT liveness:

```
POST /chain/pause
{ "reason": "CVE-2026-XXXX: mempool exploit detected",
  "signing_key": "<validator_ed25519_sk_hex>" }
```

The pause state is persisted to RocksDB and survives node restarts. This provides a zero-day response measured in seconds -- not the hours required by governance quorum systems. Any validator can resume via POST /chain/resume with valid Ed25519 authentication.

11. Aethelas Services Ecosystem

Aethelas Chain is not merely a blockchain protocol. It is the foundational settlement layer for five production-grade services that collectively address India's sovereign intelligent infrastructure gap -- from blockchain API access to decentralised storage, AI inference, physical edge compute, and distributed GPU networks.

Service	Core Proposition	Value	Market Analogue	Target ARR (2030)
Aethelas Services	Managed blockchain API, AI fraud scoring, enterprise subnets	Alchemy (Rs 85K Cr val.), Infura		Rs 4,680 Crore
Aethelas Drive	S3-compatible, DPDP-compliant, blockchain-verified Indian cloud	AWS S3 India		Rs 102 Crore
Aethelas AI	On-chain AI inference, Fraud Shield, AI Model Marketplace	Stripe Radar, Chainalysis		Rs 1,700 Crore
Aethelas Edge	500 city-level edge nodes -- LEO satellite connected	Physical infrastructure backbone		Enables all above
Aethelas Intellect	Distributed GPU/TPU -- verifiable decentralised AI	AWS SageMaker, Azure AI		Rs 1,900 Crore

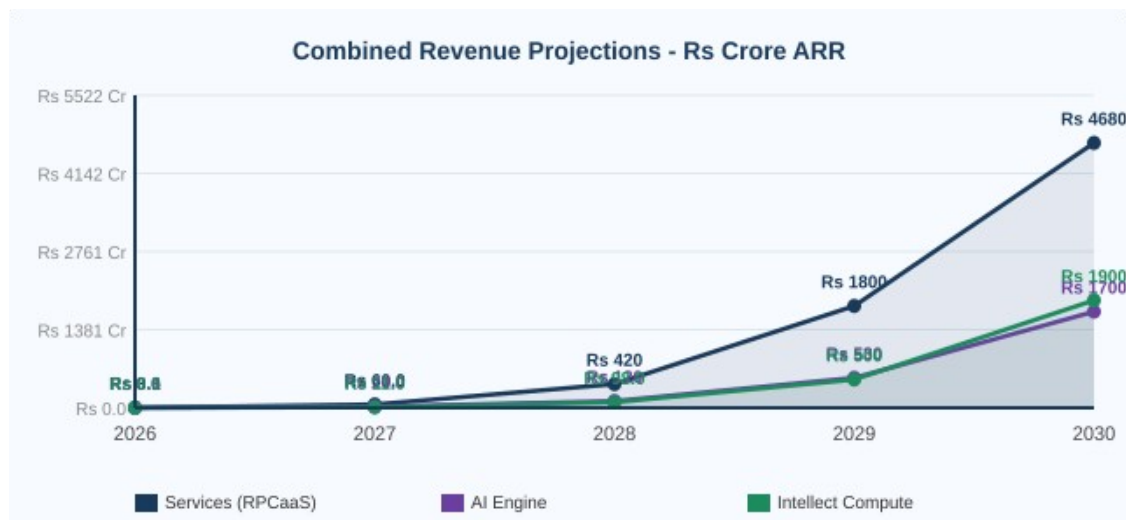


Figure 3: Combined revenue projections across Services, AI Engine, and Intellect compute -- Rs Crore ARR through 2030

12. Aethelas Services -- Blockchain-as-a-Service

Aethelas Services is the first revenue engine to launch and the most predictable cashflow business in the platform. It sells managed access to the Aethelas blockchain to developers, fintechs, and enterprises -- analogous to Alchemy and Infura for Ethereum, but with AI-native fraud scoring embedded at the API layer, a capability no competitor offers.

Alchemy (Ethereum RPC only) = Rs 85,000 Crore valuation. Infura = Rs 58,000 Crore. Aethelas Services offers RPC + AI scoring + managed nodes + enterprise subnets across the same TAM -- with a structurally differentiated product that cannot be replicated without fundamental architectural redesign.

12.1 Product Lines

Product	Description	Price Point	Launch
RPCaaS	Managed JSON-RPC + REST with JWT auth, rate limiting, billing	Rs 0-75,000/month	Phase 0 Q3 2026
Managed Validator Nodes	Dedicated nodes with 99.99% uptime SLA	Rs 8,000-1,50,000/month	Phase 0
AI Fraud Shield API	Real-time TX risk scoring -- approve/reject at mempool boundary	Rs 5,000-40,000/month	Phase 0 beta
Smart Contract Deploy	NLCC: natural language to WASM bytecode + AI security pre-audit	Rs 25,000 per deployment	Phase 1 2027
AdaptiveGas Oracle	AI fee prediction 12 blocks ahead -- Ed25519-signed, verifiable	Rs 2,000-10,000/month	Phase 1
Analytics Dashboard	Prometheus metrics, custom alerts, validator monitoring	Rs 3,000-20,000/month	Phase 1
Enterprise Private Subnet	Dedicated Aethelas chain for banks, government, regulated entities	Rs 50L-5 Cr/year	Phase 2 2028

12.2 Revenue Projections -- Services

Year	Paying Customers	Avg Monthly Revenue	ARR
2026 (H2)	100	Rs 3,000	Rs 3.6 Crore
2027	1,000	Rs 5,000	Rs 60 Crore
2028	5,000	Rs 7,000	Rs 420 Crore
2029	15,000	Rs 10,000	Rs 1,800 Crore
2030	30,000	Rs 13,000	Rs 4,680 Crore

13. Aethelas Drive -- Sovereign Cloud Storage

Aethelas Drive is India's first blockchain-verified, DPDP Act 2023 compliant cloud storage. Every file write produces an immutable on-chain proof of data localisation stored on Indian soil -- auditable by any regulator, and not subject to foreign government data requests unlike AWS and Azure which are subject to the US CLOUD Act.

Feature	Specification	Competitive Advantage
API Compatibility	S3-compatible REST API	Zero code changes for existing applications -- drop-in replacement
Redundancy	Erasure coding (RAID-6 equivalent)	Tolerates 2 simultaneous node failures -- zero data loss
Compliance	DPDP Act 2023 certificate	Blockchain-backed proof -- regulator-admissible evidence
Pricing	Rs 0.65-0.80/GB/month	Competitive with AWS S3 India -- with sovereign compliance premium
Infrastructure	Included in Edge Box (6x8TB HDD)	Zero marginal hardware cost -- storage is a byproduct of edge nodes

Year	Storage Management	Under Rate	ARR
2026	8 TB (3 pilot cities)	Rs 0.80/GB	Rs 7.7 Lakhs
2027	80 TB (30 cities)	Rs 0.80/GB	Rs 7.7 Crore
2028	267 TB (100 cities)	Rs 0.75/GB	Rs 24 Crore
2029	800 TB (300 cities)	Rs 0.70/GB	Rs 67 Crore
2030	1.3 PB (500 cities)	Rs 0.65/GB	Rs 102 Crore

14. Aethelas AI -- On-Chain Intelligence

OpenAI trained on text. Google trained on search. Aethelas trains on financial transaction intelligence -- an entirely different and enormously valuable domain. 4.8 million daily transactions x 365 days = 1.75 billion labeled financial records per year. This corpus is worth more than the code. It grows every day. It cannot be bought. It cannot be copied.

14.1 AI Products from Existing Codebase

Product	Codebase Source	Description	Market Comparable
Fraud Shield API	ai.rs verify_ai() + AIDecision	Real-time pre-mempool TX risk scoring	Stripe Radar (Rs 85,000 Cr revenue)
ContractAuditor AI	vm.rs AIVM + audit modules	AI smart contract audit in minutes vs weeks	Trail of Bits (Rs 40L per audit)
Security Mesh	consensus.rs mempool monitor +	Distributed 5-algorithm threat detection	Chainalysis (Rs 850 Cr ARR)
AdaptiveGas Oracle	AdaptiveGas engine	AI fee prediction 12 blocks ahead -- signed	Blocknative gas oracle
AI Model Marketplace	AIVM host functions, Layer 3	Developer AI models, 15-30% marketplace cut	Apple App Store (Rs 170,000 Cr/year)
Governance Advisor	AI governance layer	Economic and security impact modelling	No direct comparable

14.2 Revenue Projections -- AI Engine

Revenue Stream	2027	2028	2029	2030
Fraud Shield API	Rs 8 Cr	Rs 40 Cr	Rs 150 Cr	Rs 400 Cr
ContractAuditor AI	Rs 3 Cr	Rs 20 Cr	Rs 80 Cr	Rs 200 Cr
AI Model Marketplace	Rs 2 Cr	Rs 20 Cr	Rs 150 Cr	Rs 700 Cr
Security Mesh (enterprise)	Rs 5 Cr	Rs 30 Cr	Rs 120 Cr	Rs 300 Cr
Total AI ARR	Rs 20 Crore	Rs 120 Crore	Rs 530 Crore	Rs 1,700 Crore

15. Aethelas Edge -- Physical Infrastructure

Jio spent Rs 1,50,000 Crore to cover India with mobile internet. Aethelas covers India with city-level edge compute for Rs 750 Crore -- because the node operator model means local businesses and ISPs buy the hardware, earn AETH rewards, and Aethelas never pays for physical assets directly. Exactly how Airbnb built a hotel empire without owning a single hotel.

15.1 Edge Box Hardware Specification

Component	Specification	Cost (Rs)	Purpose
CPU	AMD EPYC 7313P -- 16 cores / 32 threads / 3.0 GHz	Rs 1,20,000	All workloads
RAM	128 GB DDR4 ECC (8x16GB)	Rs 80,000	AI inference + full blockchain state
NVMe SSD	2x4TB Samsung 980 Pro	Rs 60,000	Chain data, RocksDB, AI model weights
HDD Array	6x8TB Seagate Exos RAID-6 (~32TB usable)	Rs 1,20,000	Aethelas Drive distributed storage
GPU	NVIDIA RTX 4090 24GB or A4000 16GB	Rs 1,80,000	AIVM on-chain inference, AI Tier rewards
Satellite Terminal	OneWeb / Jio SpaceFiber LEO -- 100-400 Mbps	Rs 1,50,000	Primary WAN in non-fibre cities
Total per Node (bulk)	Bulk pricing at 100+ units	Rs 7-8.65 Lakhs	Complete edge compute stack

15.2 Node Economics

Item	Monthly (Rs)	Notes
Total Monthly Opex	Rs 46,000	Satellite + electricity + colocation + maintenance
Validator AETH rewards	Rs 50,000-2,00,000	Principal revenue stream
Drive storage fees	Rs 20,000-50,000	60% utilisation assumption
AI inference fees	Rs 30,000-1,00,000	GPU utilisation-dependent
Net Monthly Profit	Rs 64,000-3,34,000	After all operating costs

Hardware Payback Period	8-14 months	Conservative to optimistic range
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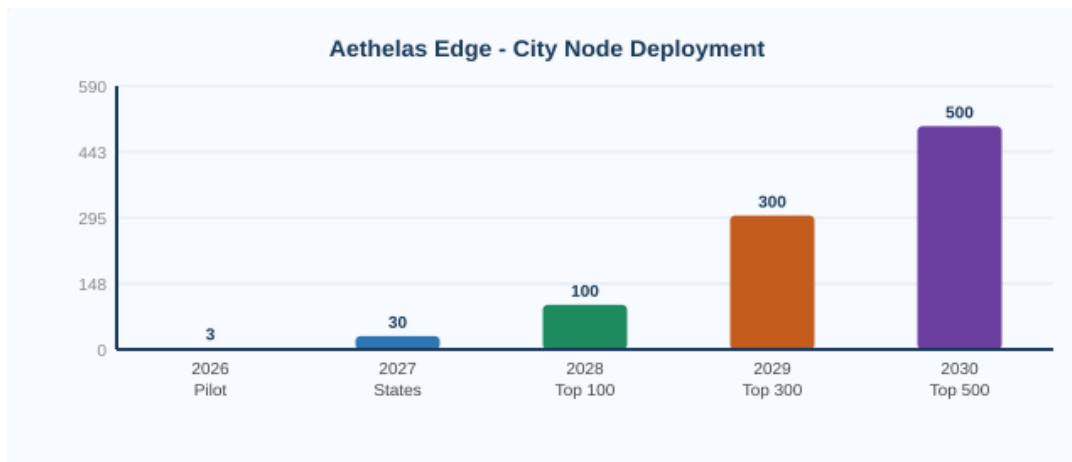


Figure 4: Aethelas Edge city node deployment -- 3 pilot cities to 500 cities across India and Southeast Asia

16. Aethelas Intellect -- Distributed AI Compute

This is what centralised AI clouds fundamentally cannot offer: trustless AI. Not 'AI on blockchain' as a marketing claim -- but genuine cryptographic verifiability of every model output, backed by BFT consensus across 500 geographically distributed inference nodes.

16.1 The Verifiable AI Advantage

Problem with Centralised AI	Aethelas Solution
Model weights are proprietary -- no independent verification	All weights stored on-chain -- any participant can inspect and audit
Outputs cannot be validated -- you must trust the provider	Every AIDecision carries Ed25519 signature + SHA-256 input_hash
Single point of failure or censorship by provider	500 BFT nodes -- no single entity can manipulate or deny service
Foreign infrastructure processes Indian data (CLOUD Act)	All inference at Indian city nodes -- DPDP Act 2023 compliant

16.2 Revenue Projections

Revenue Stream	2027	2028	2029	2030
AI inference fees (20% cut)	Rs 5 Cr	Rs 40 Cr	Rs 200 Cr	Rs 800 Cr
Enterprise AI SLA	Rs 3 Cr	Rs 25 Cr	Rs 120 Cr	Rs 400 Cr
AI training compute rental	Rs 2 Cr	Rs 15 Cr	Rs 80 Cr	Rs 300 Cr
Government AI contracts	Rs 0	Rs 10 Cr	Rs 60 Cr	Rs 250 Cr
Total Intellect ARR	Rs 11 Crore	Rs 98 Crore	Rs 500 Crore	Rs 1,900 Crore

17. Investment Thesis

17.1 The Opportunity

Aethelas addresses the intersection of three multi-trillion-dollar markets: the global AI infrastructure market (\$1T+ by 2030), the blockchain services market (\$67B by 2030), and India's sovereign digital infrastructure mandate under the DPDP Act and Digital India programme. No competitor addresses all three simultaneously with a unified physical-software-token architecture.

Phase	Period	Investment	Funding Source	Key Milestones
Phase 0 -- Seed	Q2-Q4 2026	Rs 2.23 Crore	SISFS Rs 50L + Angel	Mainnet live, 100 RPCaaS customers, 3 cities
Phase 1 -- Series A	2027	Rs 24.15 Crore	Series A -- Rs 30Cr at Rs 300Cr val.	Rs 10-20L MRR, 30 cities, AETH token launched
Phase 2 -- Series B	2028	Rs 79.69 Crore	Series B -- Rs 150Cr at Rs 800Cr val.	Rs 50-100Cr ARR, 100 cities, government contract
Phase 3 -- IPO	2029-2030	Rs 431.25 Crore	NSE/BSE IPO or Series C	Rs 1,000Cr ARR, 500 cities, audited 3 years
Phase 4 -- Global	2031-2032	Rs 1,000 Crore	Public markets	SE Asia, frontier AI, Rs 10,000 Cr ARR

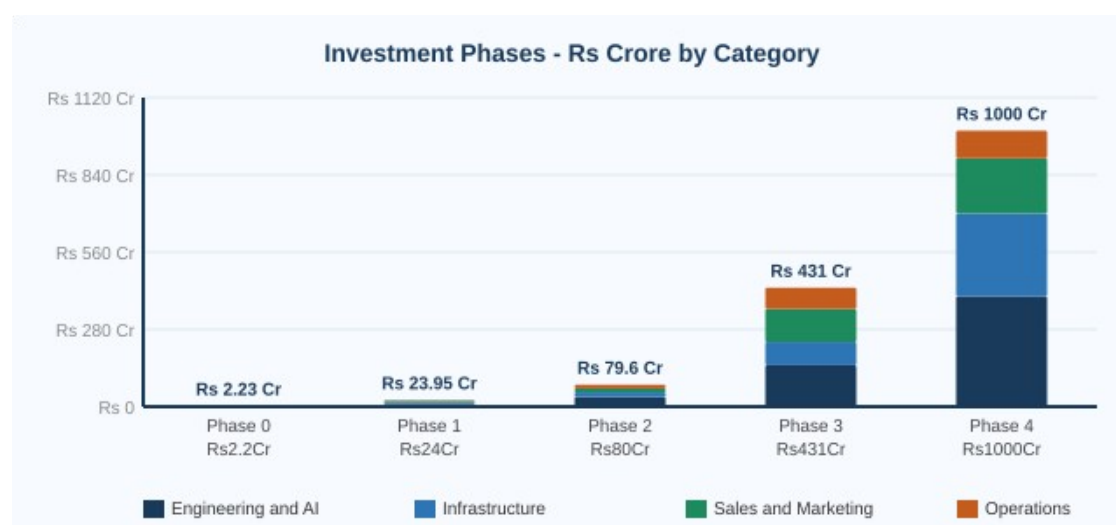


Figure 5: Investment breakdown by phase and category -- Engineering, Infrastructure, Sales, Operations

17.2 Investor Returns

Investor	Investment	2030 Valuation	Multiple	Est. IRR
SISFS (govt grant)	Rs 50 Lakhs	N/A -- non-dilutive	Grant	N/A
Angel Investor	Rs 1.5 Crore	Rs 150 Crore (10%)	100x	~120% IRR
Series A	Rs 30 Crore	Rs 3,000 Crore (10%)	100x	~90% IRR
Series B	Rs 150 Crore	Rs 5,000 Crore (10%)	33x	~60% IRR

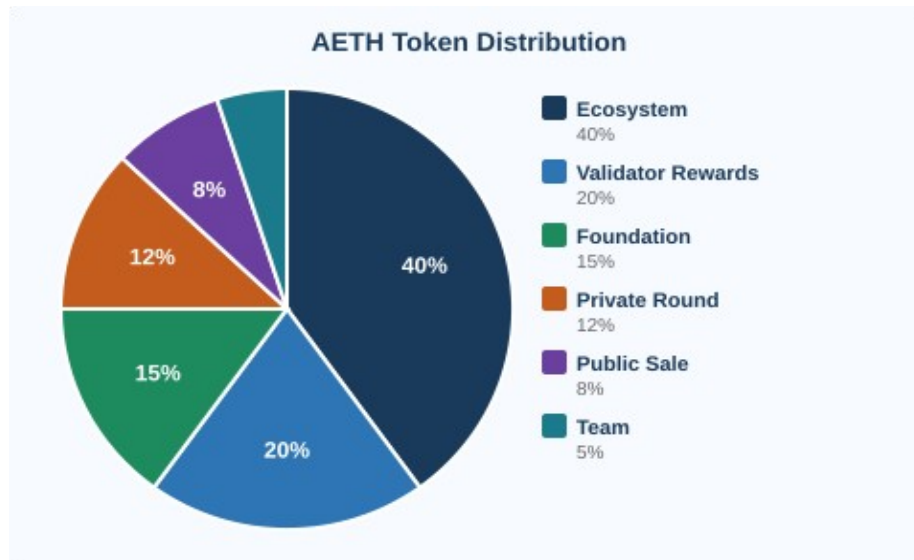


Figure 6: AETH token distribution -- Ecosystem (40%), Validator Rewards (20%), Foundation (15%), Private (12%), Public (8%), Team (5%)

18. Roadmap

Milestone	Target	Status
20-node testnet -- 20/20 nodes synced	Q2 2026	COMPLETE
84/84 unit tests passing	Q2 2026	COMPLETE
18/18 penetration tests blocked	Q2 2026	COMPLETE
On-chain governance -- 66.7% threshold	Q2 2026	COMPLETE
Emergency chain pause/resume	Q2 2026	COMPLETE
AI score persistence across restarts	Q2 2026	COMPLETE
Per-IP rate limiting + auto-ban	Q2 2026	COMPLETE
Explorer v2.0 -- 20 node + faucet + threats	Q2 2026	COMPLETE
Security audit Phase 2 (CertiK/Hacken)	Q3 2026	In Progress
50-node Hetzner cloud testnet	Q3 2026	Planned
GitHub public release	Q3 2026	Planned
RPCaaS Phase 0 -- 100 customers	Q3 2026	Planned
Edge pilot -- Patna, Mumbai, Bangalore	Q3 2026	Planned
Mainnet launch (post-audit)	Q4 2026	Planned
AETH token listing	Q4 2026	Planned
30-city Edge deployment	2027	Planned
AI Model Marketplace launch	2027	Planned
100-city Edge deployment	2028	Planned
Rs 1,000 Crore ARR -- IPO readiness	2029-2030	Target

19. Conclusion

Infrastructure is destiny. The protocols that define how data is stored, how transactions are validated, and how intelligence is distributed will shape the economic architecture of the next century. Aethelas Chain is an infrastructure bet -- not on a single application, but on the foundational layer beneath all applications and all markets.

Aethelas Chain has demonstrated -- through 84 passing unit tests, 18 blocked attack vectors, and 20 synchronised testnet nodes -- that the AI-blockchain convergence is not a theoretical proposition. It is running software. It is provable technology. It is ready.

The five service engines translate that technology into a Rs 10,000 Crore ARR business by 2032, built at 100x less capital than comparable physical infrastructure through the node operator model. The Rs 1,537 Crore total investment spans six years and six phases, beginning with a Rs 2.23 Crore seed round targeting SISFS and angel investors.

The intelligence flywheel is structural, not incremental: more transactions generate richer AI training data, which improves model performance, which improves security and efficiency, which attracts more users, which generates more transactions. This compounding advantage cannot be replicated by legacy protocols without fundamental redesign.

THE AETHELAS VISION

We envision a world where decentralised infrastructure is not just trustless, but intelligent. Not just transparent, but adaptive. Not just permissionless, but accessible to every developer, user, and institution on the planet. Infrastructure that learns, that improves, that defends itself -- and that belongs to no single nation, company, or individual.

Aethelas Chain is the foundation for that world.

Appendix: On-Chain Constants Reference

Constant	Value	Source File
CHAIN_ID	aethelas-mainnet-1	types.rs
MAX_TX_AMOUNT	10,000,000,000,000,000	types.rs
BLOCK_GAS_LIMIT	10,000,000	types.rs
BASE_BLOCK_REWARD	50,000,000 (50 AETH)	state.rs
HALVING_INTERVAL	4,200,000 blocks	state.rs
SLASH_DOUBLE_SIGN	5% of stake	state.rs
SLASH_DOWNTIME	1% of stake	state.rs
JAIL_DURATION	86,400 blocks (~24h)	state.rs
AI_SCORE_MIN	0.5	ai.rs
AI_SCORE_MAX	3.0	ai.rs
DEFAULT_MAX_PEERS	200	p2p.rs
API_RATE_GLOBAL	200 req/s	api.rs
API_RATE_PER_IP	30 req/s	api.rs
FAUCET_AMOUNT	100 AETH	api.rs
GOVERNANCE_THRESHOLD	66.67%	types.rs
UNBONDING_BLOCKS	604,800 (~7 days)	state.rs

DISCLAIMER: This white paper is for informational purposes only and does not constitute financial advice or a securities offering. Revenue projections are targets, not guarantees. All technical specifications are derived from the production codebase as of Q2 2026. Please conduct your own due diligence.

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